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Nonparametric Prior Learning in Differential Equation Modeling

Junxiong Jia
School of Mathematics and Statistics, Xi'an Jiaotong University
and
Deyu Meng, Zongben Xu
School of Mathematics and Statistics, Xi'an Jiaotong University
Pazhou Laboratory (Huangpu)
and
Fang Yao*
Department of Probability and Statistics,
Center for Statistical Science, Peking University

Abstract

This paper addresses Bayesian inference related to partial differential equations (PDEs), particularly nonparametric regression constrained by PDEs. To effectively encode prior information, we propose a novel framework that learns a prediction function of the prior distribution from historical training datasets. We introduce hyper-prior and hyper-posterior distributions and derive a generalization error estimate, which accommodates data-dependent priors by extending the concept of differential privacy. Some mild conditions are given to validate the error estimate, where various typical PDEs such as diffusion and Darcy flow equations can be integrated. We thus formulate an infinite-dimensional optimization problem to obtain the point estimate of the hyper-posterior. Numerical examples demonstrate the performance of our proposed method in learning the prediction function of priors.

Keywords: nonparametric Bayesian inference; data-dependent prior; learning prediction function; inverse problems of PDEs

*Junxiong Jia is the first author. Fang Yao is the corresponding author, E-mail: fyao@math.pku.edu.cn.
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1 Introduction

Estimating function parameters in partial differential equations (PDEs) from observed data is a fundamental problem in various scientific and engineering fields. Such problems can be framed as regression tasks constrained by PDEs. From a deterministic perspective, these inverse problems have been thoroughly studied in the mathematical literature (Engl et al. 1996). Recently, statistical approaches to efficient parameter estimation—particularly Bayesian inference—have gained significant attention (Cotter et al. 2013, Oates et al. 2019).

Unlike deterministic methods, Bayesian inference provides a general framework for addressing uncertainties in inverse problems (Arridge et al. 2019, Stuart 2010), originally developed in finite-dimensional spaces (Kaipio & Somersalo 2005). However, finite-dimensional theories often fail to characterize the continuous nature of inverse problems of PDEs. Thus, this work focuses on recent advances in Bayesian inversion for infinite-dimensional (non-parametric) spaces, as detailed in Dashti & Stuart (2017) and Stuart (2010). Bayesian inverse problems face two key challenges: constructing effective priors and developing computationally efficient algorithms. Effective priors are essential for algorithmic efficiency, acting as regularizers and encoding prior knowledge of model parameters. We now briefly summarize two research directions for constructing effective priors.

In constructing structured priors, a common approach uses convergent series expansions with coefficients from finite-dimensional distributions, yielding uniform and Gaussian priors (Dashti & Stuart 2017). These have significantly contributed to the advancement of sequential Monte Carlo algorithms (Beskos et al. 2015) and full waveform inversion (Dunlop & Yang 2021). For further insights of Gaussian measures in Bayesian inversion, see Bui-Thanh et al. (2013), Dashti et al. (2013), and referenced works. The Besov prior, valued for edge preservation (Lassas et al. 2009), often employs Haar wavelet bases, which can impose restrictive assumptions (Arridge et al. 2019). As an alternative, Markkanen et al. (2019) proposed the Cauchy difference prior. While these priors characterize the smoothness of the function parameters, they may not fully reflect specific inverse problem structures.

Optimal posterior contraction in infinite-dimensional settings presents unique challenges, as many priors fail to ensure posterior consistency (Arridge et al. 2019). Thus, prior-induced errors may persist even with infinite amount of data. For linear forward operators with Gaussian priors and noise, consistency is well-established (Knapik et al. 2016, Szabó et al. 2015), achieving minimax contraction rates when prior regularity matches the true function. In nonlinear cases, tail conditions on prior measures were proposed in Vollmer (2013), and recent studies applying measure concentration theory have derived minimax-optimal posterior contraction rates (Nickl & Söhl 2017, Nickl 2020). For detailed insights into posterior contraction in nonlinear inverse problems, see Monard et al. (2019), Nickl

(2023), and cited works. As summarized in Arridge et al. (2019), the findings on posterior consistency and optimal contraction rates underscore the regularizing characteristics of the prior measure. However, research on selecting appropriate priors for fixed data and specific noise levels remains limited.

The challenge in creating effective prior measures in infinite-dimensional spaces is highlighted due to their complexity, which is akin to manually designed features in machine learning. The paper draws a parallel with deep learning's ability to outperform traditional feature engineering by automatically extracting features from data (Goodfellow et al. 2016), suggesting a need for data-driven methods to learn prior measures. Despite numerous studies focusing on learning regularizers from data, there is a noted limitation in equating regularizers to probability measures, which hinders a direct connection to Bayesian inference (Dashti et al. 2013). A bilevel optimization framework has been proposed for learning regularizers, with a focus on parameterized regularizers that involve derivatives or filters (Haber & Tenorio 2003). This approach has been extended to non-smooth regularizers, primarily addressing optimization issues such as the existence of optimal solutions, rather than learning theory (Antil et al. 2020, Reyes et al. 2017). Recent developments in this field incorporate deep learning techniques, including generative adversarial networks (Lunz et al. 2018) and auto-encoder-based methods (Li et al. 2020). However, as noted by Afkham et al. (2021), the efficacy of estimated regularizer parameters in being optimally averaged can be diminished when dealing with highly complex distributions of model parameters.

1.1 Our contribution

In this study, we reframe the inverse problem as a constrained regression problem, influenced by forward processes like PDEs. This perspective casts model parameter estimation as a machine learning problem, treating the parameters as components of an unknown regression operator. Consequently, learning prior measures emerges as a meta-learning problem, akin to methods like model-agnostic meta-learning (Finn et al. 2017), albeit with distinct techniques and contexts. Recent extensions of the probably approximately correct (PAC)-Bayesian learning theory to meta-learning (Rothfuss et al. 2021, Rezazadeh 2022, Riou et al. 2025, Rothfuss et al. 2023) have primarily targeted data-independent priors, which capture shared hyperparameters across different learning tasks.

Inspired by Shu et al. (2023), we aim to develop a general method for learning prediction functions from historical inverse task datasets. The functions should generate prior measures tailored to a new single inverse problem. To this end, we construct a data-dependent prior learning method by rigorously extending PAC-Bayesian theory to infinite-dimensional settings. Unlike traditional machine learning, Bayesian inversion for PDEs employs a fully nonparametric (infinite-dimensional) approach (Stuart 2010, Nickl 2023), with both

unknown parameters and observed data in infinite-dimensional function spaces. Formulating suitable PAC-Bayesian theory here is challenging due to issues like absent classical density functions, PDE regularity, and singularities of Gaussian measures. Standard machine learning prior measures—such as the Gaussian $N(0, \sigma^2 I)$ with $\sigma^2 \in \mathbb{R}^+$—are ill-suited to this context. The principal contributions can be summarized as follows:

1. Inspired by the studies of empirical Bayesian approach (Szabó et al. 2015) and investigations of meta-learning (Shu et al. 2023), we introduce the prediction function of the prior probability measures, i.e., a function mapping from the data space to the space of prior measures. By generalizing the classical concept of differential privacy, we prove a general PAC-Bayesian generalization estimate for learning data-dependent prior probability measures of the inverse problems of PDEs.

2. We establish general conditions for forward PDE operators, encompassing both diffusion and Darcy flow equations. Utilizing estimations of PDEs, we derive specific generalization bounds under these conditions, thereby validating the sub-Gaussian assumption of loss functions, characterized by the negative logarithm of the likelihood function. To our knowledge, this represents a novel approach not previously explored in the realm of nonparametric Bayesian regression constrained by PDEs.

3. Although the derived estimate may not be optimal, we formulate an infinite-dimensional optimization problem that avoids the bilevel structure prevalent in prior learning studies. This approach is thus more computationally tractable. The proposed learning algorithm is rigorously defined on infinite-dimensional spaces, consistent with the principles of infinite-dimensional sampling algorithms (see Cotter et al. (2013)). We apply these algorithms to both backward diffusion (linear) and Darcy flow (nonlinear) problems. The numerical results not only validate the theoretical insights but also demonstrate the practical efficacy of the proposed methods.

The paper unfolds as follows: Section 2 consists of three parts. Subsection 2.1 introduces the general framework. Subsection 2.2 develops a PAC-Bayesian generalization bound. Subsection 2.3 presents refined bounds with data-dependent priors. Section 3 derives specific generalization bounds under general forward operator conditions. Section 4 constructs learning algorithms based on these bounds, applied to the Darcy flow problem in Section 5. Section 6 discusses potential extensions for future research. Additional analyses, detailed proofs, and code implementations are provided in the online supplement materials.

2 Learning Theory of Prior Measures

In this section, we develop PAC-Bayesian theory for the learning of data-dependent prior measures in a nonparametric Bayesian inversion approach.

2.1 Bayesian formulation

This subsection establishes the fundamental framework of the study. Consider a forward operator $\mathcal{G} : \mathcal{U} \to \tilde{\mathcal{H}}$, where $\mathcal{U}$ and $\tilde{\mathcal{H}}$ are separable Banach spaces. Data follows the model:

$$y_j = \mathcal{L}_{x_j}(\mathcal{G}(u)) + \eta_j, \quad \forall\, j = 1, \ldots, m, \tag{1}$$

with u as the model parameter, $x_j \in \mathcal{X}$ denoting measurement points or input functions, $y_j \in \mathcal{Y}$ as observed data, and η_j representing independent identically distributed (i.i.d.) noise. Introduce another separable Banach space $\mathcal{H} \subset \mathcal{Y}$, where $\mathcal{X}$ and $\mathcal{Y}$ are separable Banach spaces and $\mathcal{L}_{x_j} : \tilde{\mathcal{H}} \to \mathcal{H}$ is a bounded linear operator dependent on x_j. Define $\boldsymbol{y} = (y_1, \cdots, y_m)^T$, $\boldsymbol{\eta} = (\eta_1, \cdots, \eta_m)^T$, $\boldsymbol{x} = (x_1, \cdots, x_m)^T$, so the model (1) compacts to $\boldsymbol{y} = \mathcal{L}_{\boldsymbol{x}}\mathcal{G}(u) + \boldsymbol{\eta}$, where $\mathcal{L}_{\boldsymbol{x}}\mathcal{G}(u) := (\mathcal{L}_{x_1}(\mathcal{G}(u)), \ldots, \mathcal{L}_{x_m}(\mathcal{G}(u)))^T$. For inverse problems, assume a true parameter $u^\dagger$, leading to the forward model $\boldsymbol{y} = \mathcal{L}_{\boldsymbol{x}}\mathcal{G}(u^\dagger) + \boldsymbol{\eta}$. Detailed examples (linear and nonlinear problems) are given in Section 3 and Subsection A.7 of the supplement. In several subsequent sections, we instead focus on separable Hilbert spaces.

Throughout this work, let $\mathcal{P}(\mathcal{Y})$ denote the set of all probability measures on a separable Banach space $\mathcal{Y}$. In Bayesian inverse theory, the unknown parameter u is a random element under a prior measure $\mathbb{P} \in \mathcal{P}(\mathcal{U})$. The posterior measure $\mathbb{Q} \in \mathcal{P}(\mathcal{U})$ is defined by:

$$\frac{d\mathbb{Q}}{d\mathbb{P}}(u) = \frac{1}{Z_m} \exp\left(-\sum_{j=1}^{m} \Phi(u; y_j)\right), \tag{2}$$

where $\Phi(u; y_j)$ is the potential function from the likelihood, and Z_m is the normalizing constant. For a fixed u, data $\boldsymbol{y}$ depends on $\boldsymbol{x}$ and $\boldsymbol{\eta}$, so we also write the potential as $\Phi(u; x_j, \eta_j)$ for $j = 1, \ldots, m$. Conditions on $\mathcal{G}$ for formula (2) are reserved for Section 3, keeping fundamental concepts accessible to all readers..

We consider a joint probability measure $\mathbb{D}(dx, dy) := \mathbb{D}_1(dx)\mathbb{D}_2(x, u^\dagger, dy)$, where $\mathbb{D}_1$ and $\mathbb{D}_2$ are probability measures on $\mathcal{X}$ and $\mathcal{Y}$, respectively. Define $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$, so the random element $z := (x, y)$ satisfies $z \sim \mathbb{D} \in \mathcal{P}(\mathcal{Z})$. The dataset $S = \{z_j = (x_j, y_j)\}_{j=1}^m$ consists of i.i.d. samples from $\mathbb{D}$ (denoted $S \sim \mathbb{D}^m$). The measure $\mathbb{D}_2$ depends on the true parameter $u^\dagger$, $\mathbb{D}_1$, and the noise distribution $\mathbb{D}_3$ (a probability measure on $\mathcal{Y}$), with $\{x_j\}_{j=1}^m \sim \mathbb{D}_1$. For the Darcy flow model detailed in Section 3, the set $\{x_j\}_{j=1}^m$ are sparse observed points, often assumed evenly distributed over the domain, see Dunlop et al. (2017). Taking $\mathbb{D}_1$ as a uniform distribution ensures these points are statistically evenly spaced within the domain of the Darcy flow problem (Monard et al. 2021).

2.2 General theory

We assume n measurement datasets $\{S_1, \ldots, S_n\}$, each generated by a distinct parameter $u_i^\dagger$, such that $y_{ij} = \mathcal{L}_{x_{ij}}\mathcal{G}(u_i^\dagger) + \eta_{ij}$ for $i = 1, \ldots, n$ and $j = 1, \ldots, m_i$. Here, $\{u_i^\dagger\}_{i=1}^n$ are n background true parameters, with dataset S_i defined as $S_i = \{(x_{i1}, y_{i1}), \ldots, (x_{im_i}, y_{im_i})\}$ for all $i = 1, \ldots, n$. Further assume the background true parameters are random elements sampled from a probability measure $\mathscr{E} \in \mathcal{P}(\mathcal{U})$. This parallels the "environment" concept in meta-learning (Maurer 2005).

Observe that $\mathbb{D}(dx, dy) = \mathbb{D}_2(x, u, dy)\mathbb{D}_1(dx)$ reveals a correspondence between each $u \sim \mathscr{E}$ and a measure $\mathbb{D} \in \mathcal{P}(\mathcal{Z})$. Based on these observations, we can introduce a probability measure $\mathcal{T}$ defined on $\mathcal{P}(\mathcal{Z}) \times \mathbb{Z}^+$ such that $(\mathbb{D}, m) \sim \mathcal{T}$ corresponds to a parameter u sampled from $\mathscr{E}$ and a positive integer m sampled from some probability distribution defined on $\mathbb{Z}^+$. With these notations, the data S_i can be recognized as a sample of $\mathbb{D}_i^{m_i}$ with i.i.d. samples $(\mathbb{D}_i, m_i) \sim \mathcal{T}$ for $i = 1, \ldots, n$.

Assume the prior measure $\mathbb{P}$ is parameterized by $v \in \mathcal{V}$ (a separable Banach space), so that $\mathbb{P} = \mathbb{P}(\cdot; v)$. Define a parameterized mapping $f(\cdot; m_i, \theta) : \mathcal{Z}^{m_i} \to \mathcal{V}$ with $\theta \in \Theta$ (a separable Banach space). Since m_i is irrelevant to the following discussions, we often omit it and just use $f(\cdot; \theta)$. Here, we define $S_i \mapsto f(S_i; \theta)$ as the *prediction function* of the prior measure. For θ, we consider two probability measures: the hyper-prior $\mathscr{P} \in \mathcal{P}(\Theta)$ and hyper-posterior $\mathscr{Q} \in \mathcal{P}(\Theta)$. Our goal is to develop algorithms that integrate the information encoded in $\{S_1, \ldots, S_n\}$ with the hyper-prior $\mathscr{P}$ to produce the hyper-posterior $\mathscr{Q}$. This measure $\mathscr{Q}$ should be capable of yielding a meaningful prior measure $\mathbb{P}$.

Remark 2.1. *If $\mathcal{V} = \Theta$ and f is a constant mapping that maps every $S \in \mathcal{Z}^{m_i}$ to θ, then we can consider the prior measure $\mathbb{P} = \mathbb{P}(\cdot; \theta)$, where $\mathscr{P}$ and $\mathscr{Q}$ are the hyper-prior and hyper-posterior measures for the parameter θ, respectively.*

Remark 2.2. *Once the hyper-posterior is fixed, the data-dependent base prior $\mathbb{P}(\cdot; f(S; \theta))$ relies on a single dataset S for a newly encountered inverse problem, consistent throughout this paper. After training, parameters $\theta \sim \mathscr{Q}$ depend on all historical datasets $\{S_i\}_{i=1}^n$. Conceptually, $\mathscr{Q}$ extracts information from the data distribution $\mathbb{D}$ or environment $\mathcal{T}$.*

Remark 2.3. *For simplicity, we can model $\mathbb{P}$ as a Gaussian measure $\mathcal{N}(v, \mathcal{C}_0)$ defined on $\mathcal{U}$, parameterized by a learnable mean v and fixed covariance $\mathcal{C}_0$. To ensure equivalence across means v, $\mathcal{V}$ is taken to be the Cameron-Martin space of $\mathcal{C}_0$. The mapping f could be a neural network with parameters $\theta \in \Theta$. A sophisticated example is in Subsection 2.3.*

To validate our theoretical results, define a loss function $\ell : \mathcal{U} \times \mathcal{Z} \to \mathbb{R}$ and denote the expected error under $\mathbb{D}$ as $\mathcal{L}(u, \mathbb{D}) := \mathbb{E}_{z\sim\mathbb{D}}\ell(u, z)$. For inverse problems, since the true distributions of u and $\boldsymbol{\eta}$ are unknown, we define the empirical error $\hat{\mathcal{L}}(u, S) :=$

$\frac{1}{m}\sum_{j=1}^{m}\ell(u,z_j)$. The Gibbs error and its empirical counterpart are given by $\mathcal{L}(\mathbb{Q},\mathbb{D}) := \mathbb{E}_{u\sim\mathbb{Q}}\mathcal{L}(u,\mathbb{D})$ and $\hat{\mathcal{L}}(\mathbb{Q},S) := \mathbb{E}_{u\sim\mathbb{Q}}\hat{\mathcal{L}}(u,S)$. To distinguish hyper-prior/hyper-posterior from base prior/posterior, refer to (2) as the base prior and base posterior. Since $\mathbb{Q}$ depends on S and $\mathbb{P}$, denote it as $\mathbb{Q}(S,\mathbb{P})$ for clarity. Inspired by the investigations of meta-learning (Shu et al. 2023, Rothfuss et al. 2021), the performance of the hyper-posterior is measured via the *transfer-error*: $\mathcal{L}(\mathscr{Q},\mathcal{T}) := \mathbb{E}_{\theta\sim\mathscr{Q}}\left[\mathbb{E}_{(\mathbb{D},m)\sim\mathcal{T}}\left[\mathbb{E}_{S\sim\mathbb{D}^m}\mathcal{L}(\mathbb{Q}(S,\mathbb{P}_S^\theta),\mathbb{D})\right]\right]$, where $\mathbb{P}_S^\theta := \mathbb{P}(\cdot;f(S;\theta))$. While the transfer error is unknown in practice, we can estimate it using the empirical *multi-task error* defined by $\hat{\mathcal{L}}(\mathscr{Q},S_1,\ldots,S_n) := \mathbb{E}_{\theta\sim\mathscr{Q}}\left[\frac{1}{n}\sum_{i=1}^{n}\hat{\mathcal{L}}(\mathbb{Q}(S_i,\mathbb{P}_{S_i}^\theta),S_i)\right]$.

Let us denote $D_{\mathrm{KL}}(\cdot||\cdot)$ be the usual Kullback-Leibler (KL) divergence of two probability measures. Then, we provide an upper bound on the true transfer error $\mathcal{L}(\mathscr{Q},\mathcal{T})$ in terms of the empirical multi-task error $\hat{\mathcal{L}}(\mathscr{Q},S_1,\ldots,S_n)$ plus some general complexity terms.

Theorem 2.4. *Given the data space $\mathcal{Z}=\mathcal{X}\times\mathcal{Y}$ where $\mathcal{X}$ and $\mathcal{Y}$ are separable Banach spaces, parameter space $\mathcal{U}$ and Θ are separable Banach spaces. Assume the loss function $\ell:\mathcal{U}\times\mathcal{Z}\to\mathbb{R}$ be a measurable function. For $i=1,\ldots,n$, let $\mathbb{Q}(S_i,\mathbb{P}_{S_i}^\theta)$ be the i-th base posterior measure, $\mathscr{P}\in\mathcal{P}(\Theta)$ be some fixed hyper-prior and $\lambda,\gamma>0$. Let us define $\mathbb{E}f := \mathbb{E}_{(\mathbb{D}_1,m_1)\sim\mathcal{T}}\cdots\mathbb{E}_{(\mathbb{D}_n,m_n)\sim\mathcal{T}}\mathbb{E}_{S_1\sim\mathbb{D}_1^{m_1}}\cdots\mathbb{E}_{S_n\sim\mathbb{D}_n^{m_n}}f$ for every measurable function f. For any confidence level $\delta\in(0,1]$, we have*

$$\begin{aligned}\mathbb{P}\Bigg(\forall\mathscr{Q}\in\mathcal{P}(\Theta),\ \mathcal{L}(\mathscr{Q},\mathcal{T}) \le\ & \hat{\mathcal{L}}(\mathscr{Q},S_1,\ldots,S_n)+\left(\frac{1}{\lambda}+\frac{1}{\gamma}\right)D_{KL}(\mathscr{Q}||\mathscr{P})\\ &+\frac{1}{\gamma}\sum_{i=1}^{n}\mathbb{E}_{\theta\sim\mathscr{Q}}D_{KL}\Big(\mathbb{Q}(S_i,\mathbb{P}_{S_i}^\theta)||\mathbb{P}_{S_i}^\theta\Big)\\ &+\frac{1}{\sqrt{n}}\ln\mathbb{E}\Big[\exp\Big(\sqrt{n}\Pi^1(\gamma)+\sqrt{n}\Pi^2(\lambda)\Big)\Big]+\frac{1}{\sqrt{n}}\ln\frac{1}{\delta}\Bigg)\ge 1-\delta,\end{aligned}\tag{3}$$

where $\mathbb{P}$ is the probability with respect to the datasets $\{S_i\}_{i=1}^n$. In estimate (3), the notations

$$\Pi^1(\gamma)=\frac{1}{\gamma}\ln\mathbb{E}_{\theta\sim\mathscr{P}}\mathbb{E}_{u_1\sim\mathbb{P}_{S_1}^\theta}\cdots\mathbb{E}_{u_n\sim\mathbb{P}_{S_n}^\theta}\exp\left(\frac{\gamma}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}\left[\mathbb{E}_{z\sim\mathbb{D}_i}\ell(u_i,z)-\ell(u_i,z_{ij})\right]\right),$$

$$\Pi^2(\lambda)=\frac{1}{\lambda}\ln\mathbb{E}_{\theta\sim\mathscr{P}}\exp\left(\frac{\lambda}{n}\sum_{i=1}^{n}\left[\mathbb{E}_{(\mathbb{D},m)\sim\mathcal{T}}\mathbb{E}_{S\sim\mathbb{D}^m}\left[\mathcal{L}(\mathbb{Q}(S,\mathbb{P}_S^\theta),\mathbb{D})\right]-\mathcal{L}(\mathbb{Q}(S_i,\mathbb{P}_{S_i}^\theta),\mathbb{D}_i)\right]\right).$$

Remark 2.5. *Here we consider n i.i.d. historical datasets from n inverse tasks. For the n base inverse problems, the parameter γ is defined at the base level (not the task level), taking the form $\gamma=n\beta$, where β is from Theorem A.1 (a classical PAC-Bayes bound) in the supplementary materials. Here, each task has a different number of measurement points. We define $\bar{m}=\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\right)^{-1}$ and take $\beta=\bar{m}$ or $\sqrt{\bar{m}}$ (see Corollary 2.10). In contrast, λ is characterized at the task level. Intuitively, each task-level inverse task can be viewed as a*

data pair. Analogous to classical settings, λ may be chosen as n or $\sqrt{n}$.

In inequality (3), the term $\ln \mathbb{E}\Big[\exp\Big(\sqrt{n}\Pi^1(\gamma) + \sqrt{n}\Pi^2(\lambda)\Big)\Big]$ is a log moment generating function that quantifies how much the empirical multi-task error deviates from the transfer-error. It is crucial to derive an explicit estimate of this term to obtain a meaningful bound.

2.3 Data-dependent prior

The conventional assumption posits that prior measures are independent of observed data. However, specifying prior parameters remains challenging in many practical inverse problems, often relying on the practitioner's expertise. To mitigate this subjectivity, empirical Bayes methods adjust these parameters using dataset S (Szabó et al. 2015). Building on this idea, we extend the prior measure $\mathbb{P}_S^\theta$ to be data-dependent in Subsection 2.2. Unlike empirical Bayes, our approach trains a probabilistic model to directly map datasets to hyper-parameters of the prior, analogous to non-Bayesian methods (Shu et al. 2023).

A PAC-Bayesian learning theory for meta-learning with data-dependent priors (Liu et al. 2021) has been developed for finite-dimensional settings. The data-dependency there arises from data separation, whereas our framework introduces a prediction function (not considered in previous work). Previous studies on data-dependent priors (not for meta-learning) leverage differential privacy (Dziugaite & Roy 2018), but this approach is too restrictive for the infinite-dimensional Gaussian measures widely used in inverse problems. We address this limitation by proposing a new assumption inspired by differential privacy. The intuition of Assumption 2.6 is that discrepancies between data-dependent prior measures can be controlled, enabling the use of techniques for data-independent priors. For further discussion on differential privacy, see (Dimitrakakis et al. 2016, Dwork et al. 2015, Hall et al. 2013) and related works.

In the following, for convenience, we define an induced measure $\mathbb{D}_\mathcal{T} \in \mathcal{P}(\mathcal{Z}^m)$ as in Maurer (2005), by $\mathbb{D}_\mathcal{T}(dS) = \mathbb{E}_{(\mathbb{D},m)\sim\mathcal{T}}\left[\mathbb{D}^m(dS)\right]$. The corresponding expectation for a measurable function f on $\mathcal{Z}^m$ is then $\mathbb{E}_{S\sim\mathbb{D}_\mathcal{T}}[f] = \mathbb{E}_{(\mathbb{D},m)\sim\mathcal{T}}\left[\mathbb{E}_{S\sim\mathbb{D}^m}[f]\right]$.

Assumption 2.6. *Let $S \in \mathcal{Z}^m$, $S' \in \mathcal{Z}^{m'}$, $\theta \in \Theta$, and $u \in \mathcal{U}$. We assume that the probability measure $\mathbb{P}_S^\theta$ is absolutely continuous with respect to the measure $\mathbb{P}_{S'}^\theta$ such that*

$$\frac{d\mathbb{P}_S^\theta}{d\mathbb{P}_{S'}^\theta}(u) = \Psi(S, S', u, \theta),$$

where the function $\Psi(S, S', u, \theta)$ satisfies $\tilde{\Psi}_E := \mathbb{E}_{S'\sim\mathbb{D}_\mathcal{T}}\mathbb{E}_{u\sim\mathbb{P}_{S'}^\theta}\mathbb{E}_{S\sim\mathbb{D}_\mathcal{T}}\Psi(S, S', u, \theta)^2 \leq \Psi_E < +\infty$ with Ψ_E being a positive constant independent of m, m', and θ.

To show that Assumption 2.6 holds under certain conditions, consider a Gaussian prior $\mathbb{P} := \mathcal{N}(0, \mathcal{C}_0)$, where $\mathcal{C}_0$ is a positive-definite, symmetric trace-class operator. Let $\mathcal{U}$ be a separable Hilbert space with orthogonal basis $\{e_k\}_{k=1}^{\infty}$. Assume the prior covariance operator has spectral decomposition $\mathcal{C}_0 = \sum_{k=1}^{\infty} \lambda_k \, e_k \otimes e_k$ with $\lambda_1 \geq \lambda_2 \geq \cdots > 0$. Define $\mathcal{U}^2$ as the Hilbert scale induced by $\mathcal{C}_0$ (see Engl et al. (1996) or Subsection 3.1). Before proceeding, we state the following assumptions, which will be employed in the subsequent analysis and examples.

Assumption 2.7. *Consider the parameter space $\Theta = \Theta_1 \times \Theta_2$, with Θ_1 a separable Hilbert space and $\Theta_2 = \mathbb{R}^{N_c}$. Assume the hyper-prior measure $\mathscr{P} = \mathscr{P}_1 \otimes \mathscr{P}_2$, where $\mathscr{P}_1$ is a probability measure on Θ_1, and $\mathscr{P}_2$ is a probability measure on $\mathbb{R}^{N_c}$ with compact support $[\theta_{min}, \theta_{max}]^{N_c}$. Here, the parameters $\theta_{min}, \theta_{max} \in \mathbb{R}$ ($\theta_{min} < \theta_{max}$) are fixed real numbers.*

We assume the above Assumption 2.7 holds true and introduce the mapping $f(S;\theta) := \Big(f_m(S;\theta_1),\, \theta_2\Big) \in \mathcal{V} := \mathcal{U}^2 \times \mathbb{R}^{N_c}$. The data-dependent learned prior measure is specified as $\mathbb{P}_S^{\theta} := \mathcal{N}\Big(f_m(S;\theta_1),\, \mathcal{C}_0(\theta_2)\Big)$, where the covariance operator

$$\mathcal{C}_0(\theta_2) = \sum_{k=1}^{N_c} e^{\theta_{2k}} \, e_k \otimes e_k + \sum_{k=N_c+1}^{\infty} \lambda_k \, e_k \otimes e_k \tag{4}$$

with $N_c \in \mathbb{N}^+$ is a predefined constant. For the mean function $f_m(S;\theta_1)$, we constrain all admissible $f_m(S;\theta_1)$ to the ball $B_{\mathcal{U}^2}(R_u) \subset \mathcal{U}^2$ defined by $B_{\mathcal{U}^2}(R_u) := \Big\{ f \in \mathcal{U}^2 \,\Big|\, \|\mathcal{C}_0^{-1} f\|_{\mathcal{U}} \leq R_u \Big\}$, where $R_u > 0$ being a sufficiently large prespecified constant. Under these assumptions, taking $0 < \epsilon < \min\left\{\frac{1}{4}e^{-\theta_{\max}}, \frac{1}{4\lambda_1}\right\}$ yields the finiteness of $\tilde{\Psi}_E$:

$$\tilde{\Psi}_E \leq \Psi_E = \exp\left(\frac{(4+8\epsilon^2)R_u^2}{\epsilon}\right) \frac{N_c}{\sqrt{1-4\epsilon e^{\theta_{\max}}}} \prod_{k=N_c+1}^{\infty} (1-4\epsilon\lambda_k)^{-\frac{1}{2}} < +\infty. \tag{5}$$

Detailed explanations and the derivation of (5) are provided in Subsection A.4 of the supplementary materials. To refine Theorem 2.4, we introduce two additional assumptions.

Assumption 2.8 (sub-Gaussian assumption of Π^1)**.** *For $i = 1, \ldots, n$, we assume that the random variables $\mathcal{L}(u_i, \mathbb{D}_i) - \ell(u_i, z_i)$ are sub-Gaussian with variance factor s_I^2. Specifically, for some $\tilde{\gamma} > 0$, the following inequality holds:*

$$V_i^1 := \mathbb{E}_{(\mathbb{D}_i, m_i)\sim\mathcal{T}} \mathbb{E}_{z_i\sim\mathbb{D}_i} \mathbb{E}_{\theta\sim\mathscr{P}} \mathbb{E}_{u_i\sim\mathbb{P}_{S_i}^{\theta}} \exp\left(\tilde{\gamma}\left[\mathcal{L}(u_i, \mathbb{D}_i) - \ell(u_i, z_i)\right]\right) \leq \exp\left(\tilde{\gamma}^2 s_I^2/2\right).$$

Assumption 2.9 (sub-Gaussian assumption of Π^2)**.** *For $i = 1, \ldots, n$, we assume that the random variables $\mathbb{E}_{(\mathbb{D},m)\sim\mathcal{T}} \mathbb{E}_{S\sim\mathbb{D}^m} \mathcal{L}(\mathbb{Q}(S, \mathbb{P}^{\theta}), \mathbb{D}) - \mathcal{L}(\mathbb{Q}(S_i, \mathbb{P}_{S_i}^{\theta}), \mathbb{D}_i)$ are sub-Gaussian with*

variance factor s_{II}^2. *Specifically, for some* $\tilde{\lambda} > 0$, *the following inequality holds:*

$$V_i^2 := \mathbb{E}_{(\mathbb{D}_i, m_i)\sim\mathcal{T}} \mathbb{E}_{S_i\sim\mathbb{D}_i^{m_i}} \mathbb{E}_{\theta\sim\mathscr{P}} \exp\left(\tilde{\lambda}[\mathbb{E}_{(\mathbb{D},m)\sim\mathcal{T}} \mathbb{E}_{S\sim\mathbb{D}^m} \mathcal{L}(\mathbb{Q}(S, \mathbb{P}_{S_i}^{\theta}), \mathbb{D}\right) - \mathcal{L}(\mathbb{Q}(S_i, \mathbb{P}_{S_i}^{\theta}), \mathbb{D}_i)])$$
$$\leq \exp(\tilde{\lambda}^2 s_{II}^2/2).$$

Under Assumptions 2.6, 2.8, and 2.9, we resolve the challenge of interchanging the expectations $\mathbb{E}$ (from Theorem 2.4) and $\mathbb{E}_{u\sim\mathbb{P}_S^{\theta}}$ (with $S \sim \mathbb{D}_{\mathcal{T}}$), yielding a refined estimate.

Corollary 2.10. *Assume that all of the assumptions in Theorem 2.4, Assumptions 2.6, 2.8, and 2.9 hold true. Let us denote* $\bar{m} = \left(\frac{1}{n}\sum_{i=1}^n \frac{1}{m_i}\right)^{-1}$. *For some* $\gamma \geq 2\sqrt{n}$, $\lambda \geq 2\sqrt{n}$, *and any* $\delta \in (0, 1]$, *we have*

$$\mathbb{P}\Bigg(\forall \mathscr{Q} \in \mathcal{P}(\Theta),\ \mathcal{L}(\mathscr{Q}, \mathcal{T}) \leq \hat{\mathcal{L}}(\mathscr{Q}, S_1, \ldots, S_n) + \left(\frac{1}{\lambda} + \frac{1}{\gamma}\right) D_{KL}(\mathscr{Q}||\mathscr{P})$$
$$+ \frac{1}{\gamma}\sum_{i=1}^n \mathbb{E}_{\theta\sim\mathscr{Q}} D_{KL}\Big(\mathbb{Q}(S_i, \mathbb{P}_{S_i}^{\theta})||\mathbb{P}_{S_i}^{\theta}\Big)$$
$$+ \frac{n}{2\gamma}\ln\Psi_E + \frac{\gamma s_I^2}{n\bar{m}} + \frac{\lambda s_{II}^2}{2n} + \frac{1}{\sqrt{n}}\ln\frac{1}{\delta}\Bigg) \geq 1 - \delta.$$

Remark 2.11. *In Bayesian frameworks, the loss function is often associated with noise distribution, with Gaussian and Laplace noise as standard assumptions (Stuart 2010) yielding unbounded loss functions. We focus on sub-Gaussian assumptions for unbounded losses, contrasting with the bounded loss assumption in deterministic PDE inverse problems linked to the generalized Bayes' formula and PAC-Bayesian theory. Here, the generalized Bayes' formula refers to Bayes' formula with tempered likelihoods; see Guedj (2019), Alquier (2024) for details. Bounded loss functions are discussed in Subsection A.6 of the supplement. Another standard assumption is the sub-Gamma condition (see Supplement, Subsection A.3).*

Remark 2.12. *The parameter* γ *is typically chosen as* $\gamma = n\sqrt{\bar{m}}$ *or* $n\bar{m}$ *(where* $\bar{m}$ *is defined in Corollary 2.10). Accordingly, the external term from the data-dependent prior is* $\frac{n}{2\gamma}\ln\Psi_E = \frac{1}{2\sqrt{\bar{m}}}\ln\Psi_E$ *or* $\frac{1}{2\bar{m}}\ln\Psi_E$, *which asymptotically vanishes as* $\bar{m} \to \infty$.

Remark 2.13. *An additional term arises when the prior measure depends on dataset* S. *It appears that the bound in Corollary 2.10 is larger than that of the data-independent prior (Corollary A.6 in the supplement). However, the KL divergence term decreases when the prior depends on* S, *creating a trade-off between the KL term and the extra term* $\frac{n}{2\gamma}\ln\Psi_E$. *With* $\gamma = n\sqrt{\bar{m}}$, *see the estimate (5), the extra term scales as* $\frac{R_u^2}{\sqrt{\bar{m}}}$. *Since data-independent priors often lie far from the posterior, we can expect that introducing data-dependent priors is meaningful.*

3 Learning Priors for Inverse Problems

This section presents our general framework for linear and nonlinear inverse problems, designed to unify the general theory with typical PDE problems. Here, we set $\gamma = n\beta$ throughout. For details on β, see Subsections A.2 and A.3 in the supplement.

3.1 General assumptions and formulas

We outline standard prior and noise measure assumptions in nonparametric Bayesian inversion (Stuart 2010, Cotter et al. 2013). The recent Gaussian process work (Mora et al. 2025) may also apply; see Subsection A.5 of the supplementary materials.

Assumption 3.1. *Let us give some basic assumptions used in the rest of this section:*

1. *Assume $\mathcal{U}$ is an infinite-dimensional separable Hilbert space. The prior measure $\mathbb{P}_S^\theta := \mathcal{N}(f_m(S;\theta_1), \mathcal{C}_0(\theta_2))$ defined on $\mathcal{U}$ as in Subsection 2.3, with the mean function $f_m(S;\theta_1)$ lying in the Cameron-Martin space of $\mathcal{C}_0$ (see Subsection 2.3) for all $S \in \mathcal{Z}^m$ and $\theta \in \Theta$. Without loss of generality, let $\{\lambda_k, e_k\}_{k=1}^\infty$ be the eigensystem of $\mathcal{C}_0$, where $\{\lambda_k\}_{k=1}^\infty$ are non-increasing and $\{e_k\}_{k=1}^\infty$ are normalized eigenfunctions. Assume there exists $s_0 \in [0,1)$ such that $\mathcal{C}_0^s$ is trace class for all $s > s_0$.*

2. *Assume that the space $\mathcal{H}$ introduced in Subsection 2.1 is either $\mathbb{R}^{N_d}$, where N_d is a positive integer, or $\mathcal{H}$ is an infinite-dimensional separable Hilbert space.*

3. *Assume that the noise η is distributed according to a Gaussian distribution $\mathbb{D}_3$ defined as $\mathbb{D}_3 := \mathcal{N}(0,\Gamma)$, where $\Gamma : \mathcal{H} \to \mathcal{H}$ is a self-adjoint, positive-definite linear operator, but not necessarily trace class. Similar settings can be found in (Agapiou et al. 2013).*

4. *If $\mathcal{H}$ is an infinite-dimensional Hilbert space, let us introduce $\mathcal{C}_1 : \mathcal{H} \to \mathcal{H}$, which is a self-adjoint, positive-definite, trace class, linear operator. Suppose $\mathcal{C}_1\Gamma = \Gamma\mathcal{C}_1$ and there exists $s_1 \in [0,1)$ such that $\mathcal{C}_1^s$ is trace class for all $s > s_1$.*

Consider $\mathcal{H}$, a finite or infinite-dimensional space. For the infinite-dimensional case, we introduce the Hilbert scale (Engl et al. 1996), central to our analysis. Since $\mathcal{C}_1$ (Assumption 3.1) is injective and self-adjoint, $\mathcal{H} = \overline{\mathcal{R}(\mathcal{C}_1)} \oplus \mathcal{R}(\mathcal{C}_1)^\perp = \overline{\mathcal{R}(\mathcal{C}_1)}$, making $\mathcal{C}_1^{-1} : \mathcal{R}(\mathcal{C}_1) \to \mathcal{H}$ a densely defined, unbounded, symmetric, positive-definite linear operator. It extends to a self-adjoint operator with domain $\mathcal{D}(\mathcal{C}_1^{-1}) := \{u \in \mathcal{H} \mid \mathcal{C}_1^{-1}u \in \mathcal{H}\}$. Thus, we define the Hilbert scale $(\mathcal{H}^t)_{t\in\mathbb{R}}$, where $\mathcal{H}^t := \overline{\mathcal{M}_1}^{\|\cdot\|_{\mathcal{H}^t}}, \mathcal{M}_1 := \bigcap_{\ell=0}^\infty \mathcal{D}(\mathcal{C}_1^{-\ell})$, with inner product and norm $\langle u, v\rangle_{\mathcal{H}^t} = \left\langle \mathcal{C}_1^{-t/2}u, \mathcal{C}_1^{-t/2}v\right\rangle_{\mathcal{H}}, \|u\|_{\mathcal{H}^t} = \left\|\mathcal{C}_1^{-t/2}u\right\|_{\mathcal{H}}$. Assumption 3.1 states $\mathcal{C}_1\Gamma = \Gamma\mathcal{C}_1$ (infinite-dimensional case), a harmless assumption since $\mathcal{C}_1$ only defines the function space. Analogously, define the Hilbert scale $(\mathcal{U}^t)_{t\in\mathbb{R}}$ for $\mathcal{U}$ and prior covariance $\mathcal{C}_0$: $\mathcal{U}^t := \overline{\mathcal{M}_2}^{\|\cdot\|_{\mathcal{U}^t}}, \mathcal{M}_2 := \bigcap_{\ell=0}^\infty \mathcal{D}(\mathcal{C}_0^{-\ell})$, with $\langle u, v\rangle_{\mathcal{U}^t} = \left\langle \mathcal{C}_0^{-t/2}u, \mathcal{C}_0^{-t/2}v\right\rangle_{\mathcal{U}}, \|u\|_{\mathcal{U}^t} = \left\|\mathcal{C}_0^{-t/2}u\right\|_{\mathcal{U}}$. The Hilbert scale motivates the following assumptions.

Assumption 3.2. *Suppose constants $s \in (s_1, 1]$, $\tilde{s} \in (s_0, 1]$, and $\alpha \geq 0$ such that*

1. *If the space $\mathcal{H}$ is a separable Hilbert space, for all $w \in \mathcal{H}^{\alpha-s}$, $s \in (s_1, 1]$ and $\rho \in [\lceil \alpha - s_1 - 1 \rceil, \alpha - s_1)$, we assume $\left\| \mathcal{C}_1^{\frac{s}{2}} \Gamma^{-\frac{1}{2}} w \right\|_{\mathcal{H}} \leq C_1 \left\| \mathcal{C}_1^{\frac{s-\alpha}{2}} w \right\|_{\mathcal{H}}$ and $\left\| \mathcal{C}_1^{-\frac{\rho}{2}} \Gamma^{\frac{1}{2}} w \right\|_{\mathcal{H}} \leq C_2 \left\| \mathcal{C}_1^{\frac{\alpha-\rho}{2}} w \right\|_{\mathcal{H}}$.*
2. *If the space $\mathcal{H}$ is a separable Hilbert space, for any $x \in \mathcal{X}$ and $u, u_1, u_2 \in \mathcal{U}^{1-\tilde{s}}$, we have*

$$\begin{aligned} \|\mathcal{C}_1^{-\frac{s}{2}} \Gamma^{-\frac{1}{2}} \mathcal{L}_x \mathcal{G}(u)\|_{\mathcal{H}} &\leq M_1(\|u\|_{\mathcal{U}^{1-\tilde{s}}}), \\ \|\mathcal{C}_1^{-\frac{s}{2}} \Gamma^{-\frac{1}{2}} (\mathcal{L}_x \mathcal{G}(u_1) - \mathcal{L}_x \mathcal{G}(u_2))\|_{\mathcal{H}} &\leq M_2(\|u_1\|_{\mathcal{U}^{1-\tilde{s}}}, \|u_2\|_{\mathcal{U}^{1-\tilde{s}}}) \|u_1 - u_2\|_{\mathcal{U}^{1-\tilde{s}}}, \\ \|\mathcal{C}_1^{-\frac{s}{2}} \Gamma^{-\frac{1}{2}} (\mathcal{L}_x \mathcal{G}(u) - \mathcal{L}_{x'} \mathcal{G}(u))\|_{\mathcal{H}} &\to 0, \quad \text{as } x \to x' \text{ in } \mathcal{X}, \end{aligned}$$

where $M_1(\cdot)$ is a monotonic non-decreasing function and $M_2(\cdot, \cdot)$ is a function monotonic non-decreasing separately in each argument. If $\mathcal{H}$ is a finite-dimensional space, say $\mathcal{H} = \mathbb{R}^{N_d}$, we can omit the operator $\mathcal{C}_1$ from the aforementioned requirements.

The two inequalities given in (1) of Assumption 3.2 can be intuitively interpreted as $\Gamma \simeq \mathcal{C}_1^{\alpha}$, where the notation $\simeq$ indicates that the two operators are equal in some sense of norm equivalence. Employing the second inequality in (1) of Assumption 3.2, we obtain $\mathbb{E}_{\eta \sim \mathbb{D}_3} \|\eta\|_{\mathcal{H}^{\alpha-s}}^2 \leq C_2^2 \mathbb{E}_{\eta \sim \mathbb{D}_3} \|\mathcal{C}_1^{\frac{s}{2}} \Gamma^{-\frac{1}{2}} \eta\|_{\mathcal{H}}^2 < +\infty$ by utilizing Lemma 3.3 of Agapiou et al. (2013). This implies that the noise η drawn from $\mathbb{D}_3$ almost surely belongs to $\mathcal{Y} := \mathcal{H}^{\alpha-s}$.

When $\mathcal{H}$ is infinite-dimensional, we define the loss function as $\ell(u, z) := \Phi(u; x, y) = \frac{1}{2} \|\Gamma^{-\frac{1}{2}} \mathcal{L}_x \mathcal{G}(u)\|_{\mathcal{H}}^2 - \langle \Gamma^{-\frac{1}{2}} y, \Gamma^{-\frac{1}{2}} \mathcal{L}_x \mathcal{G}(u) \rangle_{\mathcal{H}}$. Under the assumed Gaussian noise model, this function acts as the potential, or equivalently, the negative log-likelihood; see, e.g., (Dashti & Stuart 2017, Stuart 2010). To establish generalization estimates, it is essential to verify Assumptions 2.8 and 2.9 in the current setting. This requires a rigorous formulation the generalized posterior measure $\mathbb{Q}(S, \mathbb{P}_S^{\theta})$ associated with the base inverse problem.

Theorem 3.3. *Suppose that Assumptions 2.7, 3.1, and 3.2 are satisfied. For the function $M_1(\cdot)$ in Assumption 3.2, assume there exists a constant $\delta_0 > 0$ such that for all $\delta \in [0, \delta_0)$, the quantity $\mathbb{E}_{u \sim \mathbb{P}_S^{\theta}} \exp\left(\delta M_1 \left(\|u\|_{\mathcal{U}^{1-\tilde{s}}}\right)^2\right)$ is finite. Let $\mathcal{H}$ be a separable Hilbert space. For any fixed $\theta \in \Theta$, suppose that $\|f_m(S; \theta_1)\|_{\mathcal{U}^{1+\tilde{s}}} \leq M(r)$ whenever $\|y_j\|_{\mathcal{H}^{\alpha-s}} \leq r$ for all $j = 1, \ldots, m$, where $M(\cdot)$ is a non-decreasing function. Additionally, assume that $\|f_m(S; \theta_1) - f(S'; \theta_1)\|_{\mathcal{U}^{1+\tilde{s}}} \to 0$, as $x_j \to x_j'$ in $\mathcal{X}$ and $y_j \to y_j'$ in $\mathcal{H}^{\alpha-s}$, for $j = 1, \ldots, m$, where $S' = \{(x_j', y_j')\}_{j=1}^m$. Under these conditions, the generalized posterior measure $\mathbb{Q}(S, \mathbb{P}_S^{\theta})$ is absolutely continuous with respect to $\mathbb{P}_S^{\theta}$, and*

$$\frac{d\mathbb{Q}(S, \mathbb{P}_S^{\theta})}{d\mathbb{P}_S^{\theta}}(u) = \frac{1}{Z_m} \exp\left(-\frac{\beta}{m} \sum_{j=1}^{m} \Phi(u; x_j, y_j)\right), \tag{6}$$

where $\beta > 0$ *is a constant,* $\Phi(u; x_j, y_j) := \frac{1}{2}\left\|\Gamma^{-\frac{1}{2}}\mathcal{L}_{x_j}\mathcal{G}(u)\right\|_{\mathcal{H}}^2 - \left\langle \Gamma^{-\frac{1}{2}} y_j, \Gamma^{-\frac{1}{2}}\mathcal{L}_{x_j}\mathcal{G}(u)\right\rangle_{\mathcal{H}}$, *and* $Z_m \in (0, \infty)$ *is the normalizing constant. Moreover, the mapping* $S \mapsto \mathbb{Q}(S, \mathbb{P}_S^\theta)$ *is continuous with respect to the Hellinger metric:* $d_{Hell}\Big(\mathbb{Q}(S, \mathbb{P}_S^\theta), \mathbb{Q}(S', \mathbb{P}_{S'}^\theta)\Big) \to 0$, *as* $x_j \to x_j'$ *in* $\mathcal{X}$ *and* $y_j \to y_j'$ *in* $\mathcal{H}^{\alpha-s}$. *If* $\mathcal{H} = \mathbb{R}^{N_d}$ *is finite-dimensional, then all of the above results remain valid upon replacing* $\|\cdot\|_{\mathcal{H}^{\alpha-s}}$ *with* $\|\cdot\|_{\mathcal{H}}$, *which corresponds to the standard Euclidean norm on* $\mathbb{R}^{N_d}$.

Remark 3.4. *If the space* $\mathcal{H}$ *is finite-dimensional, we define the potential function as* $\Phi(u; x_j, y_j) := \frac{1}{2}\left\|\Gamma^{-\frac{1}{2}}(\mathcal{L}_{x_j}\mathcal{G}(u) - y_j)\right\|_{\mathcal{H}}^2$, *which is non-negative. Therefore, the condition* $\mathbb{E}_{u\sim\mathbb{P}_S^\theta} \exp\left(\delta M_1 \left(\|u\|_{\mathcal{U}^{1-\tilde{s}}}\right)^2\right) < \infty$ *is not required to ensure the lower bound in the proof.*

Remark 3.5. *For nonlinear inverse problems, assume the true parameter* $u^\dagger$ *lies in the ball* $B_{\tilde{s}}(R_u) := \{u \in \mathcal{U} \mid \|u\|_{\mathcal{U}^{1-\tilde{s}}} \le R_u\}$ *(see Subsection 3.2). Similar constraints are imposed in deterministic and statistical theories (Engl et al. 1996, Nickl 2023, 2020). Define* $\mathbb{P}_{R_u}^{S,\theta}$ *as the truncated Gaussian measure:* $\mathbb{P}_{R_u}^{S,\theta}(du) := \frac{1}{\mathbb{P}_S^\theta(B_{\tilde{s}}(R_u))} 1_{B_{\tilde{s}}(R_u)}(u)\mathbb{P}_S^\theta(du)$. *When* $u^\dagger \in B_{\tilde{s}}(R_u)$ *is known a priori, using this truncated prior is more reasonable (Beskos et al. 2018, Rudolf & Sprungk 2018); see Subsections A.4 and A.6 in the supplement for additional discussions. Based on an analogous illustration of Theorem 3.3, we find that*

$$\frac{d\mathbb{Q}(S,\mathbb{P}_{R_u}^{S,\theta})}{d\mathbb{P}_{R_u}^{S,\theta}}(u) = \frac{\mathbb{P}_S^\theta(B_{\tilde{s}}(R_u))}{Z_m^{R_u}} \exp\left(-\frac{\beta}{m}\sum_{j=1}^m \Phi(u; x_j, y_j)\right) 1_{B_{\tilde{s}}(R_u)}(u) \tag{7}$$

is well-defined, where $Z_m^{R_u}$ *is the normalization constant.*

3.2 Linear and nonlinear problems

In this subsection, assume the forward operator $\mathcal{G}$ is linear. Recall notations from Subsection 2.1: denote $\boldsymbol{y} = (y_1, \ldots, y_m)^T$ and $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_m)^T$. The forward problem with m measurements compacts to $\boldsymbol{y} = \mathcal{L}_{\boldsymbol{x}}\mathcal{G}u + \boldsymbol{\eta}$, where $\mathcal{L}_{\boldsymbol{x}}\mathcal{G} = (\mathcal{L}_{x_1}\mathcal{G}, \ldots, \mathcal{L}_{x_m}\mathcal{G})^T$. The noise vector $\boldsymbol{\eta} \sim \mathcal{N}(0, \Gamma_m)$ with $\Gamma_m = \mathrm{diag}(\Gamma, \ldots, \Gamma)$. The generalized posterior measure in Theorem 3.3 thus takes the form:

$$\frac{d\mathbb{Q}(S,\mathbb{P}_S^\theta)}{d\mathbb{P}_S^\theta}(u) \propto \exp\left(-\frac{\beta}{2m}\|\mathcal{L}_{\boldsymbol{x}}\mathcal{G}u\|_{\Gamma_m}^2 + \frac{\beta}{m}\langle \mathcal{L}_{\boldsymbol{x}}\mathcal{G}u, \boldsymbol{y}\rangle_{\Gamma_m}\right), \tag{8}$$

where $\|\mathcal{L}_{\boldsymbol{x}}\mathcal{G}u\|_{\Gamma_m}^2 = \sum_{j=1}^m \|\Gamma^{-\frac{1}{2}}\mathcal{L}_{x_j}\mathcal{G}u\|_{\mathcal{H}}^2$ and $\langle \mathcal{L}_{\boldsymbol{x}}\mathcal{G}u, \boldsymbol{y}\rangle_{\Gamma_m} = \sum_{j=1}^m \langle \Gamma^{-\frac{1}{2}}\mathcal{L}_{x_j}\mathcal{G}u, \Gamma^{-\frac{1}{2}}y_j\rangle_{\mathcal{H}}$. Since the forward operator is assumed to be linear, the generalized posterior measure

$\mathbb{Q}(S, \mathbb{P}_S^\theta)$ is actually a Gaussian measure $\mathcal{N}(u_p, \mathcal{C}_p)$, Example 6.23 in Stuart (2010) with

$$u_p = f_m(S;\theta_1) + \mathcal{C}_0(\theta_2)(\mathcal{L}_{\boldsymbol{x}}\mathcal{G})^*\Big(\frac{m}{\beta}\Gamma_m + \mathcal{L}_{\boldsymbol{x}}\mathcal{G}\mathcal{C}_0(\theta_2)(\mathcal{L}_{\boldsymbol{x}}\mathcal{G})^*\Big)^{-1}(\boldsymbol{y} - \mathcal{L}_{\boldsymbol{x}}\mathcal{G}f_m(S;\theta_1)), \quad (9)$$

$$\mathcal{C}_p = \mathcal{C}_0(\theta_2) - \mathcal{C}_0(\theta_2)(\mathcal{L}_{\boldsymbol{x}}\mathcal{G})^*\Big(\frac{m}{\beta}\Gamma_m + \mathcal{L}_{\boldsymbol{x}}\mathcal{G}\mathcal{C}_0(\theta_2)(\mathcal{L}_{\boldsymbol{x}}\mathcal{G})^*\Big)^{-1}\mathcal{L}_{\boldsymbol{x}}\mathcal{G}\mathcal{C}_0(\theta_2). \quad (10)$$

In Lemma 3.6, we show that Assumptions 2.8 and 2.9 hold under mild PDE-related conditions. This validates applying the general theory in Section 2 to the linear case.

Lemma 3.6. *Assume Assumptions 2.7, 3.1, and 3.2 hold. For the linear forward operator, reformulate the conditions as* $\|\mathcal{C}_1^{-s/2}\Gamma^{-1/2}\mathcal{L}_x\mathcal{G}u\|_{\mathcal{H}} \le M_1\|u\|_{\mathcal{U}^{1-\tilde{s}}}$ *and* $\|\mathcal{C}_1^{-s/2}\Gamma^{-1/2}\mathcal{L}_x\mathcal{G}(u_1 - u_2)\|_{\mathcal{H}} \le M_2\|u_1 - u_2\|_{\mathcal{U}^{1-\tilde{s}}}$, *with positive constants* M_1, M_2. *Let the probability measure* $\mathscr{E}$ *have compact support* $\mathrm{supp}\,\mathscr{E} \subset B_{\tilde{s}}(R_u) := \{u \in \mathcal{U} : \|u\|_{\mathcal{U}^{1-\tilde{s}}} \le R_u\}$ *for some* $R_u \in \mathbb{R}^+$. *Define* $\tilde{C} := \|\mathcal{C}_1^{s/2}\|_{\mathcal{B}(\mathcal{H})}$ *(the operator norm) and assume* $\max\left\{3\tilde{\gamma}\tilde{C}^2M_2^2, 3\tilde{\gamma}^2M_1^2\,Tr(\mathcal{C}_1^s)\right\} \le \min\left\{e^{-\theta_{max}}, \lambda_1^{-1}\right\}$, *where* θ_{max} *and* λ_1 *are as in Assumption 3.1. The base posterior measure follows from (8). Consequently,* s_I^2 *and* s_{II}^2 *in Assumptions 2.8 and 2.9 are finite positive reals for* $\tilde{\gamma} := \frac{\gamma}{n\bar{m}}$ *and* $\tilde{\lambda} := \frac{\lambda}{n}$, *with* γ *and* λ *are proportional to* $n\bar{m}$ *and* n *respectively. Under some boundedness assumptions (include the backward diffusion problem),* s_I^2 *and* s_{II}^2 *become independent of* $\tilde{\gamma}$ *and* $\tilde{\lambda}$, *allowing* $\gamma := n\sqrt{\bar{m}}$ *and* $\lambda := 2\sqrt{n}$.

For brevity, we omit the explicit formulas for s_{I} and s_{II} and the auxiliary boundedness assumptions, which are detailed in Subsection B.1 of the online supplement. We now focus on nonlinear forward operators. Unlike linear cases, the generalized posterior measure lacks a tractable explicit form, rendering the exact expectation intractable—a key challenge in nonlinear settings. Leveraging Remark 3.5, we employ a truncated Gaussian prior, yielding a well-posed generalized posterior. Our proofs require the prior to be supported on a ball in some function space. Conceptually, the priors used in Nickl (2020) are also applicable here. Specifically, we state a lemma that estimates V_i^1 and V_i^2 under Assumptions 2.8 and 2.9 for nonlinear problems, validating the general theory's application.

Lemma 3.7. *Suppose that the probability measure* $\mathscr{E}$ *and the loss function* ℓ *satisfy the conditions stated in Lemma 3.6. We further assume that Assumptions 2.7, 3.1, and 3.2 are valid, with the prior measure* $\mathbb{P}_S^\theta = \mathcal{N}(f_m(S;\theta_1), \mathcal{C}_0(\theta_2))$ *being replaced by the truncated Gaussian assumption. Specifically,* $\mathbb{P}_S^\theta$ *is substituted with* $\mathbb{P}_{R_u}^{S,\theta}$ *as defined in Remark 3.5. Additionally, we assume that the base posterior measure is obtained through the generalized Bayes' formula with truncated Gaussian prior as defined in (7). Consequently, for* s_I^2 *and* s_{II}^2 *as defined in Assumptions 2.8 and 2.9, we have* $s_I^2 = 4\tilde{C}^4M_2(R_u, R_u)^4R_u^4 + \frac{1}{4}\tilde{C}^4M_1(R_u)^4R_u^4 + M_1(R_u)^2\,Tr(\mathcal{C}_1^s)R_u^2 < +\infty$, *and* $s_{II}^2 = 25\|\mathcal{C}_1^{\frac{s}{2}}\|_{\mathcal{B}(\mathcal{H})}^4M_1(R_u)^4 < +\infty$, *where* $\tilde{C} = \|\mathcal{C}_1^{\frac{s}{2}}\|_{\mathcal{B}(\mathcal{H})}$.

For concrete linear and nonlinear inverse problems, the conditions in Assumption 3.2

must be validated via PDE estimations in suitable function spaces. This allows explicit identification of constants s_{I} and s_{II}, enabling derivation of an explicit PAC-Bayesian estimate by integrating results from Section 2. Examples for backward diffusion (linear) and Darcy flow (nonlinear) problems are detailed in Subsection A.7 of the online supplement.

4 Learning Algorithms

In this section, we develop practical learning algorithms based on the derived PAC-Bayesian generalization bound. Following the finite-dimensional approach in (Rothfuss et al. 2023), we set $\gamma = n\beta$, where β is the parameter in the generalized Bayes' formula of Theorem 3.3. Next, we denote $\{Z_{m_i} = Z_{m_i}(S_i, \mathbb{P}^{\theta}_{S_i})\}_{i=1}^{n}$, noting that Z_{m_i} depends on the dataset and prior measure. With these choices, we further refine the PAC-Bayesian estimate to obtain

$$\begin{aligned}\mathcal{L}(\mathscr{Q}, \mathcal{T}) \leq & -\frac{1}{n\beta}\sum_{i=1}^{n}\mathbb{E}_{\theta\sim\mathscr{Q}}\ln Z_{m_i}(S_i, \mathbb{P}^{\theta}_{S_i}) + \frac{\lambda + n\beta}{\lambda n\beta}D_{\mathrm{KL}}(\mathscr{Q}||\mathscr{P}) \\ & + \frac{1}{2\beta}\ln\Psi_E + \frac{\beta s_{\mathrm{I}}^2}{\bar{m}} + \frac{\lambda s_{\mathrm{II}}^2}{2n} + \frac{1}{\sqrt{n}}\ln\frac{1}{\delta}.\end{aligned} \tag{11}$$

To minimize the right-hand side of (11), we address the following optimization problem:

$$\arg\min_{\mathscr{Q}\in\mathcal{P}(\Theta)} \mathbb{E}_{\theta\sim\mathscr{Q}}\left[-\frac{\lambda}{\lambda + n\beta}\sum_{i=1}^{n}\ln Z_{m_i}(S_i, \mathbb{P}^{\theta}_{S_i})\right] + D_{\mathrm{KL}}(\mathscr{Q}||\mathscr{P}). \tag{12}$$

In light of this optimization problem, we aim to establish a theorem that provides an analytical solution for the optimal measure $\mathscr{Q}$, thereby avoiding the complex bilevel optimization.

Theorem 4.1. *For both bounded and unbounded loss cases, assume Assumptions 2.7, 3.1, 3.2, and the conditions in Theorem 3.3 hold. Further posit that for any dataset S and $\theta_1 \in \Theta_1$, the mean function $f_m(S;\theta_1)$ satisfies $\|f_m(S;\theta_1)\|_{\mathcal{U}^{1+\bar{s}}} \leq R_u$ for some positive constant R_u. The optimization problem (12) then has a solution defined by*

$$\frac{d\mathscr{Q}}{d\mathscr{P}}(\theta) = \frac{1}{Z_p}\exp\left(\frac{\lambda}{\lambda + n\beta}\sum_{i=1}^{n}\ln Z_{m_i}(S_i, \mathbb{P}^{\theta}_{S_i})\right), \tag{13}$$

where $Z_p := \int_{\Theta}\exp\left(\frac{\lambda}{\lambda+n\beta}\sum_{i=1}^{n}\ln Z_{m_i}(S_i, \mathbb{P}^{\theta}_{S_i})\right)\mathscr{P}(d\theta)$.

Different from the parametric setting in Rothfuss et al. (2021), the above Bayes' formula (13) does not trivially hold true. Equipped with the formula (13), we can now extract information about the parameter θ from the hyper-posterior measure $\mathscr{Q}$. Note that the

term inside the exponential of formula (13) can be calculated approximately by:

$$\begin{aligned}\sum_{i=1}^{n} \ln Z_{m_i}(S_i, \mathbb{P}^{\theta}_{S_i}) &= \sum_{i=1}^{n} \ln \int_{\mathcal{U}} \exp\left(-\frac{\beta}{m_i} \sum_{j=1}^{m_i} \Phi(u; z_{ij}) \right) \mathbb{P}^{\theta}_{S_i}(du) \\ &\approx \sum_{i=1}^{n} \ln \left[\sum_{\ell=1}^{L} \exp\left(-\frac{\beta}{m_i} \sum_{j=1}^{m_i} \Phi(u^i_\ell; z_{ij}) \right) \right] - n \ln L,\end{aligned} \tag{14}$$

where $\{u^i_\ell\}_{\ell=1}^{L}$ are samples from the base prior $\mathbb{P}^{\theta}_{S_i}$ for $i = 1, \ldots, n$. To evaluate the last term in (14), we must compute nL forward PDEs, which is computationally intensive. Hence, we focus on the maximum a posteriori (MAP) estimate in this work.

Given the parameter θ in a general separable Hilbert space Θ, we adopt the weak maximum a posteriori (MAP) estimator introduced in Agapiou et al. (2018). Let the hyper-prior $\mathscr{P} = \mathscr{P}_1 \otimes \mathscr{P}_2$ (see Assumption 2.7), where $\mathscr{P}_1$ denotes a Gaussian measure with Cameron–Martin space E_{Θ_1} and $p_2(\cdot)$ represents the density of $\mathscr{P}_2$. Under suitable conditions (see Subsection A.8 of the supplementary materials for a rigorous exposition), computing the MAP estimate reduces to solving:

$$\arg\min_{\theta} -\frac{\lambda}{\lambda + n\beta} \sum_{i=1}^{n} \ln \left[\sum_{\ell=1}^{L} \exp\left(-\frac{\beta}{m_i} \sum_{j=1}^{m_i} \Phi(u^i_\ell; z_{ij}) \right) \right] + \frac{1}{2} \|\theta_1\|^2_{E_{\Theta_1}} - \ln p_2(\theta_2). \tag{15}$$

The above minimization problem becomes computationally intensive for large n. We address this by substituting the first term with a mini-batch approximation during iteration:

$$-\frac{\lambda n}{(\lambda + n\beta) H} \sum_{h=1}^{H} \ln \left[\sum_{\ell=1}^{L} \exp\left(-\frac{\beta}{m_i} \sum_{j=1}^{m_i} \Phi(u^h_\ell; z_{hj}) \right) \right], \tag{16}$$

where the datasets $\{S_h\}_{h=1}^{H}$ are randomly selected from $\{S_i\}_{i=1}^{n}$. Algorithm 1 summarizes the resulting generic prior distribution learning procedure. For further details and a discussion on potential sampling methods, refer to Subsection A.8 of the supplementary materials.

5 Numerical Examples

We focus on the inverse problem of the Darcy flow equation, a more challenging nonlinear inverse problem than the backward diffusion problem. Numerical results for the backward diffusion problem are deferred to Subsection A.9 of the supplementary materials.

Let $\Omega = (0, 1)^2$, and define the Hilbert space $\mathcal{U}$ as $(L^2(\Omega), \langle \cdot, \cdot \rangle, \| \cdot \|)$. The Darcy flow equation takes the following form:

$$-\nabla \cdot (e^u \nabla w) = f \quad \text{in } \Omega, \qquad w = 0 \quad \text{on } \partial\Omega, \tag{17}$$

Algorithm 1 Prior Learning Algorithm

1: **Input**: Specify the hyper-prior $\mathscr{P}$, datasets $\{S_1, \ldots, S_n\}$, maximum iterative number N, the mini-batch size H, and the learning rate schedule α_k. Initialize the parameters of the prior $\mathbb{P}_\cdot^\theta = \mathcal{N}(f(\cdot;\theta_1), \mathcal{C}_0(\theta_2))$ with $\theta = \theta^0$ and set the counter k to 0;

2: **Repeat**

3: Set $k = k + 1$;

4: Select $\{S_h\}_{h=1}^H$ randomly from $\{S_i\}_{i=1}^n$ with $S_i = \{z_{i1}, \ldots, z_{im_i}\}$;

5: Calculate the parameters $\theta^{k+1} = \theta^k - \alpha_k \nabla_{\theta^k} L(\theta^k)$ with

$$L(\theta^k) = -\frac{\lambda n}{(\lambda + n\beta)H} \sum_{h=1}^{H} \ln \left[\sum_{\ell=1}^{L} \exp\left(-\frac{\beta}{m_i} \sum_{j=1}^{m_i} \Phi(u_\ell^h(\theta^k); z_{hj}) \right) \right] + \frac{1}{2} \|\theta_1^k\|_{E_{\Theta_1}}^2 - \ln p_2(\theta_2^k).$$

Variants of SGD can also be employed in this step.

6: **Until**: $k = K_{\max}$.

7: **Output**: The learned prior measure $\mathbb{P}_\cdot^{\theta^{K_{\max}}} = \mathcal{N}(f(\cdot;\theta_1^{K_{\max}}), \mathcal{C}_0(\theta_2^{K_{\max}}))$.

where f represents the sources, and $e^{u(x)}$ describes the permeability of the porous medium. For the inverse problem, the dataset consists of discrete measured values of w, obtained via the measurement operator $\mathcal{L}_{x_j}(w) = w(x_j)$ where $x_j \in \Omega$ for $j = 1, \ldots, m$. The function $u(x)$ is the parameter that needs to be inferred. As noted in Remark 2.5, we assume $m_i = m$ for $i = 1, \ldots, n$ and set $\beta = m$, $\gamma = nm$, and $\lambda = n$. For this problem, we have $\tilde{\mathcal{H}} = H^1(\Omega)$, $\mathcal{H} = \mathbb{R}$, and assume Gaussian noise $\eta \sim \mathcal{N}(0, \Gamma)$ with $\Gamma = \tau^2 I$ ($\tau > 0$). In all numerical examples, τ is set to $0.01 \max_j\{|w(x_j)|\}$. The equation is discretized on Ω using the finite element method with continuous Lagrange basis functions $\{\phi_k\}_{k=1}^N$. We define the finite-dimensional space $V_N = \text{span}\{\phi_1, \ldots, \phi_N\}$, where approximate functions $f = \sum_{k=1}^N f_k \phi_k \in V_N \subset L^2(\Omega)$. For the parameters H and L in Algorithm 1, we set $H = 20$ and $L = 10$. More details can be found in Subsection A.9 of the supplementary materials.

Recall the operator $\mathcal{C}_0$ introduced in Subsection 2.3, we take $\mathcal{C}_0 = (I - 0.1\Delta)^{-2}$, where Δ is the Laplace operator satisfying homogeneous Neumann boundary condition. For the definitions of $\theta := (\theta_1, \theta_2)$ and $\mathcal{C}_0(\theta_2)$, see Subsection 2.3. The dimension of θ_2 is $N_c = 45$ in our implementation. Here, we consider three types of base prior measures as follows: (1) Unlearned prior: $\mathbb{P} := \mathcal{N}(0, \mathcal{C}_0)$; (2) Learned data-independent prior: $\mathbb{P}^\theta := \mathcal{N}(f_m(\theta_1), \mathcal{C}_0(\theta_2))$; (3) Learned data-dependent prior: $\mathbb{P}_S^\theta := \mathcal{N}(f_m(S;\theta_1), \mathcal{C}_0(\theta_2))$.

In the above, the data-independent learned mean function $f_m(\theta_1) := \sum_{k=1}^N \theta_{1k} \varphi_k$ with $\{\phi_k\}_{k=1}^N$ as the basis of V_N. To keep the discretization-invariant property of the infinite-dimensional formulation, the data-dependent learned mean function $f_m(S;\theta_1)$ is taken to be a Fourier neural operator illustrated in Kovachki et al. (2021). Since the dataset S consists of discrete vectors, the Fourier neural network cannot be directly applied. Actually, we transform the dataset S into a function via the adjoint measurement operator $\mathcal{L}_{\boldsymbol{x}}^* : \mathcal{H} \to \tilde{\mathcal{H}}$.

For more implementation specifics, see Subsection A.9 in the supplementary materials.

When the base prior is $\mathbb{P}^\theta$, we take the hyper-prior $\mathscr{P} := \mathcal{N}(0, A_h^{-2}) \otimes \mathcal{N}(\ln \boldsymbol{\lambda}, 10 I_{N_c})$, where $A_h = 0.01I - 0.1\Delta$, $\ln \boldsymbol{\lambda} := (\ln \lambda_1, \ldots, \ln \lambda_{N_c})$, and I_{N_c} is the identity operator on $\mathbb{R}^{N_c}$. For the mean function $f_m(S; \theta_1)$, specifying a suitable hyper-prior is challenging due to the lack of intuition regarding the Fourier neural operator's parameter θ_1. Instead, we leverage the a priori regularity of $f_m(S; \theta_1)$, randomly selecting n_S datasets ($n_S = 4$ in practice), and define $g(\theta)$ as the sum of $f_m(S_i; \theta_1)$ over these datasets. Consequently, we establish $\mathscr{P}$ as $\mathscr{P} := g_\# \mathcal{N}(0, A^{-2}) \otimes \mathcal{N}(\ln \boldsymbol{\lambda}, 10 I_{N_c})$ when the base prior is $\mathbb{P}_S^\theta$.

5.1 Simple environment

We evaluate our method in both simple and complex environments. In the simple environment, the measure $\mathscr{E}$ generates ground-truth parameters via the random function:

$$u(x_1, x_2) = u_1(x_1, x_2) + u_2(x_1, x_2) + u_3(x_1, x_2), \tag{18}$$

where $u_i(x_1, x_2) = a_{i3}(1 - x_1^2)^{a_{i1}}(1 - x_2^2)^{a_{i2}} e^{-a_{i4}(x_1 - a_{i5})^2 - a_{i6}(x_2 - a_{i7})^2}$ with $i = 1, 2, 3$. For the parameters in the above formula, we let $a_{i1} \sim U(0.1, 0.5)$, $a_{i2} \sim U(0.1, 0.5)$, $a_{i3} \sim U(3, 4)$, $a_{i4} \sim U(30, 35)$, $a_{i6} \sim U(30, 35)$, $a_{15} \sim U(0.15, 0.25)$, $a_{17} \sim U(0.65, 0.75)$, $a_{25} \sim U(0.45, 0.55)$, $a_{27} \sim U(0.45, 0.55)$, $a_{35} \sim U(0.65, 0.75)$, and $a_{37} \sim U(0.15, 0.25)$, where $i = 1, 2, 3$. We generate 2000 and 50 random functions via formula (18) to construct the historical datasets $\{S_i\}_{i=1}^{2000}$ and testing datasets, respectively. For constructing these datasets, we select 100 statistically equally-spaced sparse measurement points ($m = 100$).

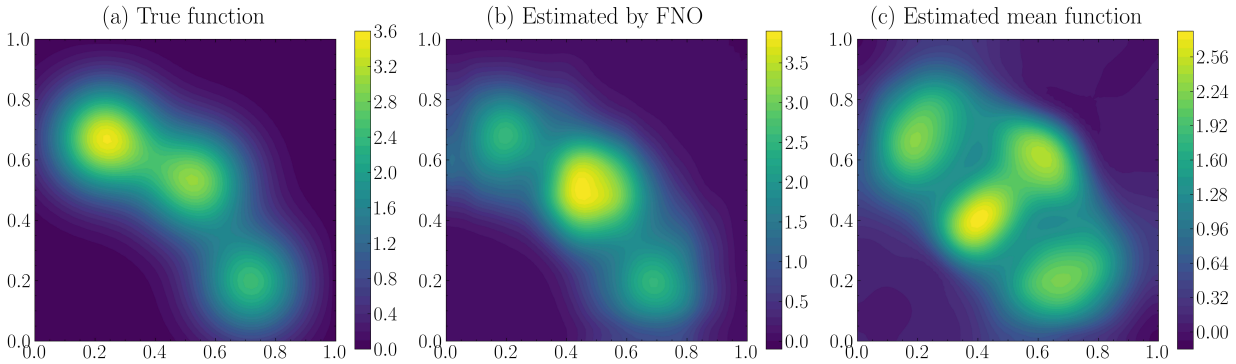


Figure 1: *True function and estimated mean functions in the simple environment setting. (a) One of the functions in the testing dataset; (b) The learned mean function $f_m(S; \theta_1)$ of $\mathbb{P}_S^\theta$; (c) The learned mean function $f_m(\theta_1)$ of $\mathbb{P}^\theta$.*

Before presenting quantitative comparisons, we provide a visual comparison in Fig. 1. In Fig. 1 (a), we display one of the functions in the testing datasets, offering an intuitive understanding of the model parameters. In Fig. 1 (b), we present the learned mean function

$f_m(S;\theta_1)$ of $\mathbb{P}_S^\theta$. Lastly, in Fig. 1 (c), we display the learned mean function $f_m(\theta_1)$ of $\mathbb{P}^\theta$. Notably, the learned mean function shown in (b) of Fig. 1 closely resembles the true underlying function. In contrast, the mean function of the learned data-independent prior depicted in (c) of Fig. 1 only represents the averaged information and visually deviates from this specific true underlying function.

The objective of learning prior measures is to enhance efficiency in solving novel inverse problems associated with historical inverse tasks. To show the proposed method's superiority, we evaluate base posteriors induced by three priors $\mathbb{P}$, $\mathbb{P}^\theta$, and $\mathbb{P}_S^\theta$ via maximum a posteriori (MAP) estimates, posterior means, and credible regions. The matrix-free Newton conjugate gradient algorithm (Ghattas & Willcox 2021) computes MAP estimates for testing datasets, while the mixture Gaussian sequential Monte Carlo (MGSMC) algorithm (Lu et al. 2024) generates samples from the base posterior. Algorithmic details are provided in Subsection A.9 of the supplementary materials.

In Table 1, we report relative errors (in L^2-norm) of the computed MAP estimates against ground truth, indexed by the number of iterations. Regardless of the environment, the learned data-dependent Bayesian model yields the smallest relative error, requiring only 5 iterations. In the simple environment, the learned data-independent model converges faster than the unlearned Bayesian model but yields a similar relative error.

Table 1: For the 50 test datasets, we list the average relative errors of the MAP estimates from base posteriors under priors $\mathbb{P}$, $\mathbb{P}^\theta$, and $\mathbb{P}_S^\theta$, respectively. Here, ItN denotes the iteration count of the inexact matrix-free Newton-conjugate gradient method.

	Simple Environment			Complex Environment		
ItN	$\mathbb{P}$	$\mathbb{P}^\theta$	$\mathbb{P}_S^\theta$	$\mathbb{P}$	$\mathbb{P}^\theta$	$\mathbb{P}_S^\theta$
1	.7322	.1932	**.0973**	.7130	.9120	**.1078**
5	.2756	.1674	**.0911**	.3515	.7056	**.0895**
10	.1614	.1574	**.0911**	.2798	.6794	**.0875**
15	.1460	.1520	**.0914**	.2714	.6723	**.0874**
20	.1401	.1457	**.0914**	.2683	.6685	**.0869**

In the Bayesian framework, we need to characterize the behavior of the base posterior. We use the MGSMC algorithm to generate 1000 samples from the posterior. Specifically, similar to other sequential Monte Carlo algorithms, the MGSMC algorithm decomposes the original problem into a series of simpler sampling problems. This decomposition relies on a sequence of positive numbers $\{h_k\}_{k=1}^{K_{\text{total}}}$ (termed temperatures) satisfying $\sum_{k=1}^{K_{\text{total}}} h_k = 1$. The temperatures $\{h_k\}_{k=1}^{K_{\text{total}}}$ are automatically determined based on the effective sample size (see supplementary materials for details). With other parameters fixed, K_{total} governs the sampling speed of the MGSMC algorithm. The left panel of Fig. 2 plots the accumulated

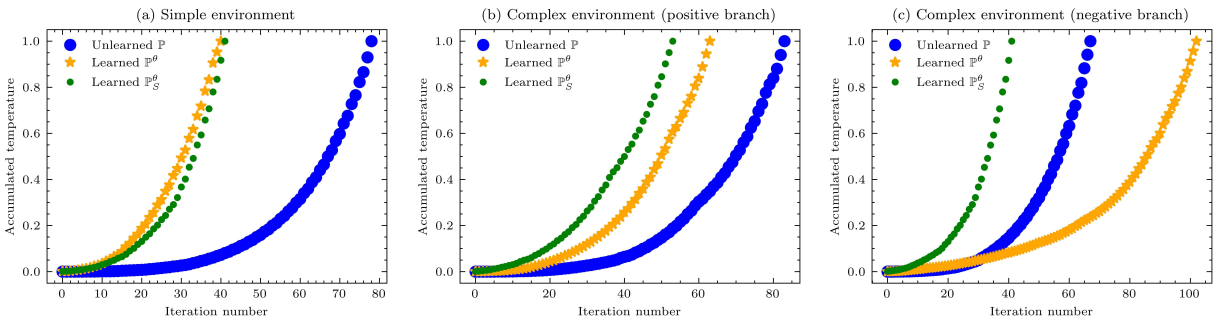


Figure 2: *Accumulated temperatures selected by effective sample size are shown in (a)-(c) for one test dataset from each case: the simple environment, the positive-valued branch, and the negative-valued branch of the complex environment. In all panels: blue dots represent unlearned Bayesian model temperatures, orange star dots denote learned data-independent Bayesian model temperatures, and green dots signify learned data-dependent Bayesian model temperatures.*

temperatures $\sum_{k=1}^{K} h_k$ $(K = 1, \ldots, K_{\text{total}})$ under the simple environment setting. The learned data-dependent and learned data-independent models achieve a similar sampling speed, and the unlearned model is significantly slower than the other two methods. Thus, the Bayesian model with learned prior substantially accelerates posterior calculation.

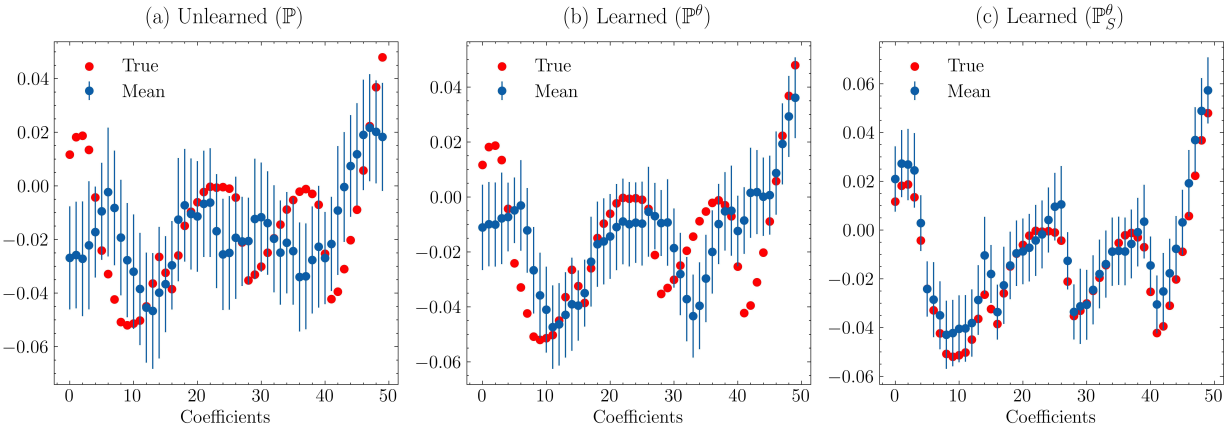


Figure 3: *In all panels, red and blue dots denote the true function's projected coefficients and the base posterior mean's projected coefficients, respectively. Blue vertical lines mark* 95% *credible intervals. Panels (a), (b), and (c) display results of simple environment under the priors* $\mathbb{P}$, $\mathbb{P}^{\theta}$, *and* $\mathbb{P}^{\theta}_{S}$, *respectively.*

In addition to sampling speed, we compare uncertainty quantification results. Since parameter u is a function, we project it onto the first 200 eigenfunctions $\{e_k\}_{k=1}^{200}$ of operator $\mathcal{C}_0$. We then compute the 95% credible intervals for the projected variables $\{\langle u, e_k \rangle\}_{k=1}^{200}$. In Fig. 3, red and blue dots denote the projected coefficients of the true function and the posterior mean of the base posterior, respectively. For clarity, we display only 50 projected points. Results from the unlearned, learned data-independent, and data-dependent models are shown in the left, middle, and right panels, respectively. Notably, all true values

lie within the 95% credible intervals obtained by the learned data-dependent model. By contrast, some true values fall outside the 95% credible intervals of the other two models. Furthermore, the credible intervals of the unlearned and learned data-independent models are wider than those of the learned data-dependent model.

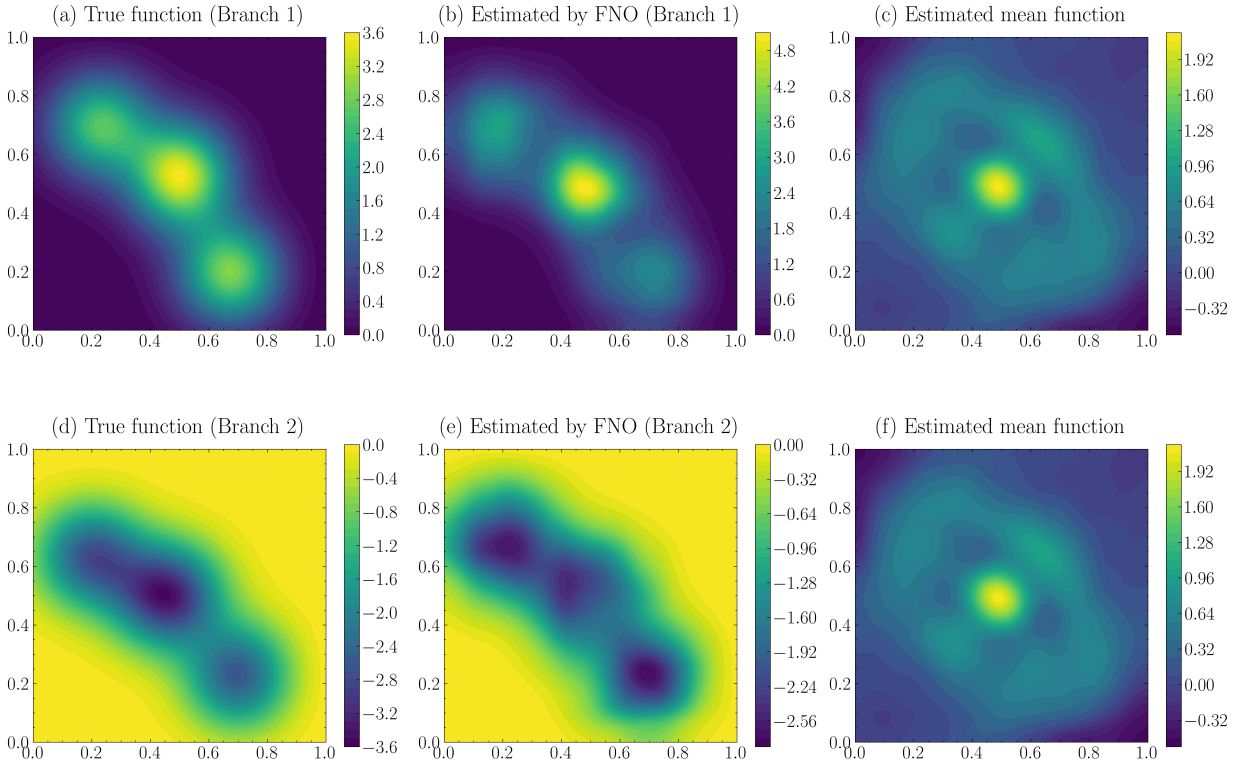


Figure 4: *True and estimated functions for Branches 1 and 2: (a) shows a random function of Branch 1; (b) and (c) present the mean functions of priors $\mathbb{P}_S^\theta$ and $\mathbb{P}^\theta$ for Branch 1, respectively; (d) depicts a random function of Branch 2; (e) and (f) display the corresponding mean functions from both priors for Branch 2.*

5.2 Complex environment

Except for the simple environment, we design a more challenging complex environment, which has two branches compared to the simple environment. Specifically, the background true functions are generated based on the formula: $u(x_1, x_2) = (2\alpha - 1)(u_1(x_1, x_2) + u_2(x_1, x_2) + u_3(x_1, x_2))$, where $\alpha \sim \text{Bern}(0.5)$ and settings are the same as in the simple environment. As in the simple environment, we generate 2000 training and 50 testing functions. In Figs. 4 (a) and (d), we show two functions randomly selected from the two branches of the test examples. Figs. 4 (b) and (e) present the learned mean functions under the data-dependent prior assumption. Notably, the Fourier neural operator distinguishes between the two branches and provides reasonable estimates. Figs. 4 (c) and (f) (same figure) display the learned mean functions under the data-independent prior assumption, demonstrating that this approach fails to capture the branches' characteristics.

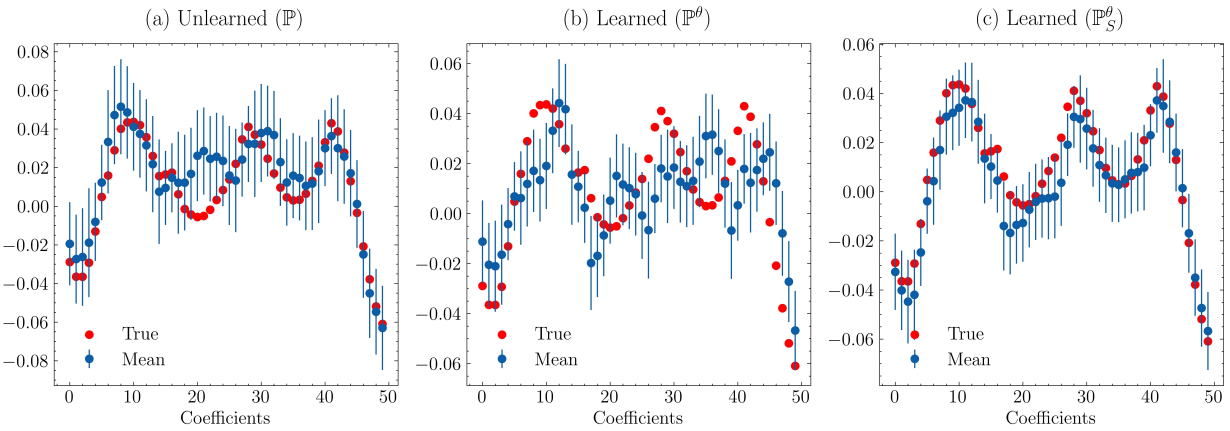


Figure 5: *In all panels, red and blue dots denote the true function's projected coefficients and the base posterior mean's projected coefficients, respectively. Blue vertical lines mark* 95% *credible intervals. Panels (a), (b), and (c) show results for a negative-valued branch (complex environment) using priors* $\mathbb{P}$, $\mathbb{P}^\theta$, *and* $\mathbb{P}_S^\theta$, *respectively.*

Table 1 shows the average relative errors of MAP estimates in the complex environment. The learned data-dependent Bayesian model yields the lowest errors and requires only 5 iterations. The learned data-independent model performs significantly worse than the other two, with detailed analysis in Subsection A.9 of the supplement. Sampling speeds are shown in Fig. 2 (b,c). For the positive-valued branch, all models exhibit similar speeds to the simple environment. For the negative-valued branch, however, the learned data-independent model requires $K_{\text{total}} > 100$—more than twice that of the data-dependent model — highlighting the efficiency of the data-dependent prior model in complex scenarios. In Fig. 5, we show the 95% credible intervals of the three methods. The results clearly demonstrate that the data-dependent prior is capable of extracting valuable information from historical inverse tasks, utilizing only the historical measurement datasets without the true parameter.

Finally, we analyze computational times. All programs ran on a system with an Intel(R) Xeon(R) Platinum 8180 CPU, 48 GB NVIDIA RTX A6000 GPU, and Ubuntu 22.04.5 LTS OS. We randomly selected a dataset generated from a negative-valued function in a complex environment and recorded the runtime of the MGSMC algorithm. The unlearned, learned data-independent, and learned data-dependent models consumed approximately 3074s, 6257s, and 1682s, respectively. Notably, the MGSMC algorithm used 1000 particles; increasing the particle number exacerbates the performance gap. By leveraging a mixture Gaussian approximation, MGSMC reduces computational cost significantly compared to the classical SMC algorithm (Dashti & Stuart 2017), though at the cost of reduced accuracy. For comparison, the classical SMC (500 particles) with preconditioned Crank–Nicolson sampling (200 steps) required approximately 57396s, 142843s, and 31362s for the respective models. This translates to a savings of nearly 7.2 hours per inverse problem for learned data-dependent models over unlearned models. Given computational constraints, we

used 500 particles; extrapolating, 2000 particles would save nearly 28.8 hours. Training $\mathbb{P}_S^\theta = \mathcal{N}(f_m(S;\theta_1), \mathcal{C}_0(\theta_2))$ with parameters $L = 10$, $H = 20$, and 7200 maximum iterations required 75750s (approximately 21 hours). While substantial, this is a one-time cost: once trained, the neural operator predictor can be employed to solve new inverse problems (given a new dataset S, no further training is needed).

6 Concluding Remarks

This paper formulates infinite-dimensional inverse problems as regression tasks constrained by forward operators (e.g., heat or elliptic equations). Leveraging meta-learning, we develop a PAC-Bayesian framework for predicting prior probability measures in infinite-dimensional spaces. A key innovation is introducing data-dependent base priors via a condition inspired by the concept of differential privacy. By extending infinite-dimensional Bayesian theory, we derive general forward operator conditions compatible with both linear and nonlinear problems. Specific generalization bounds are established through careful log-moment generating function estimations. This theory yields an optimization problem avoiding the bilevel structure typical of prior learning. Finally, we propose a tractable learning algorithm that constructs effective prior-generating mappings.

Designing neural operator architectures to process discrete datasets S and probability measures is critical for solving inverse problems. Architectural inspirations can be drawn from DeepSets (Zaheer et al. 2017) and NIO frameworks (Molinaro et al. 2023), which offer structural paradigms for handling such input modalities. While only empirical PAC-Bayes bounds are obtained in this work, deriving oracle PAC-Bayes bounds and tighter empirical bounds (Alquier 2024, Riou et al. 2025) remains an important future research direction.

7 Disclosure statement

The authors report there are no competing interests to declare.

8 Data Availability Statement

The generated data (6.84GB) have been made available at the following URL: https://drive.google.com/drive/folders/1ekb_fWj4p7CZEBsx6BOMSkb5eCBMU_9A?usp=drive_link.

SUPPLEMENTARY MATERIAL

Learn_Prior_Supp: The additional theoretical results related to Sections 2 and 3, more numerical results, all proofs are deferred to the Supplementary Material. (.pdf file).

Code and Data: Python code to generate the synthetic datasets and reproduce the simulations and corresponding output.

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